

Week 11 (CS 418 @ UIC)

Week 11 Slide Deck

Recommendation Systems; A/B Testing

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Demo Content Slide

Placeholder content for Week 11.

- Recommendation systems
- A/B testing
- Evaluation metrics

Examples of Recommender Systems

Amazon: “Customers Who Bought This...”

- Popularized **item-to-item collaborative filtering**: recommend items similar to what you bought, where “similar” means *bought by the same people*¹
- Works from behavior (purchases, views) — no ratings required
- The *Touching the Void* story is this system finding the long tail

1. Linden, Smith & York, “Amazon.com recommendations: item-to-item collaborative filtering,” *IEEE Internet Computing* 7:1 (2003).

Feeds

- **TikTok / Shorts / Reels** — feed algorithms are essentially recommenders

And Also...

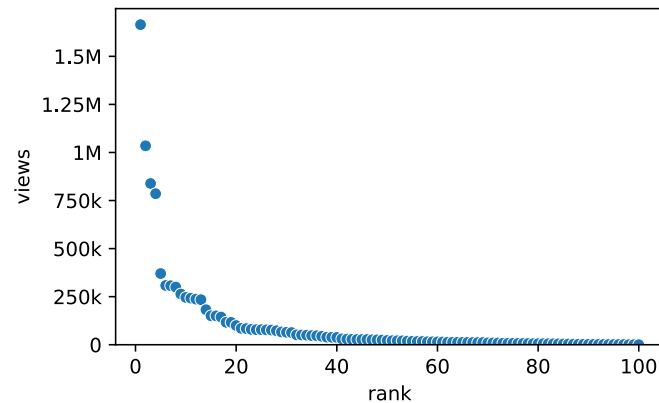
- **Spotify** — E.g. Discover Weekly, collaborative filtering over listening histories
- **Online dating, job boards, app stores** — same basic matrix structure, different items
- **News feeds** — articles similar to what you've read, or read by similar readers

The Premise

- Suppose we pulled 100 random movies from IMDB/Letterboxd
- Sort them by popularity
- Views/popularity on the y axis
- “Hits” appear on the left side

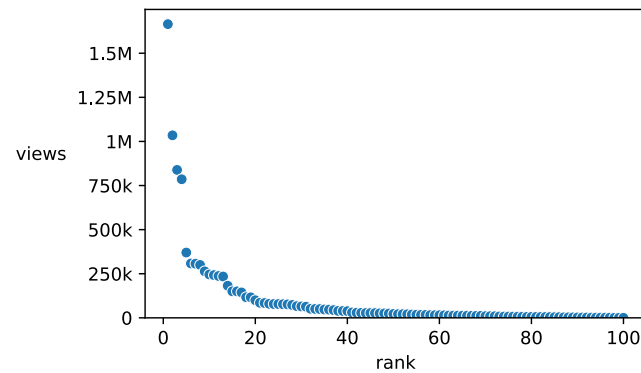
The Premise, Plotted: Synthetic Data

```
1 import numpy as np, polars as pl, seaborn as sns
2 from matplotlib.ticker import EngFormatter
3 views = np.sort(np.random.default_rng(418).lognormal(10, 2, 100))[:-1]
4 movies = pl.DataFrame({"rank": np.arange(1, 101), "views": views})
5 ax = sns.scatterplot(movies, x="rank", y="views")
6 ax.set_ylim(bottom=0)
7 ax.yaxis.set_major_formatter(EngFormatter(sep=""))
```



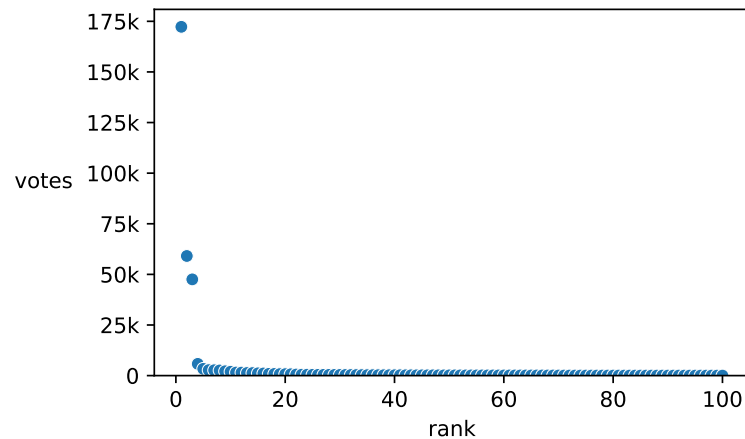
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6 ax.set_ylim(bottom=0)
7 ax.set_ylabel("views", rotation=0, labelpad=20)
8 ax.yaxis.set_major_formatter(EngFormatter(sep=""))
```



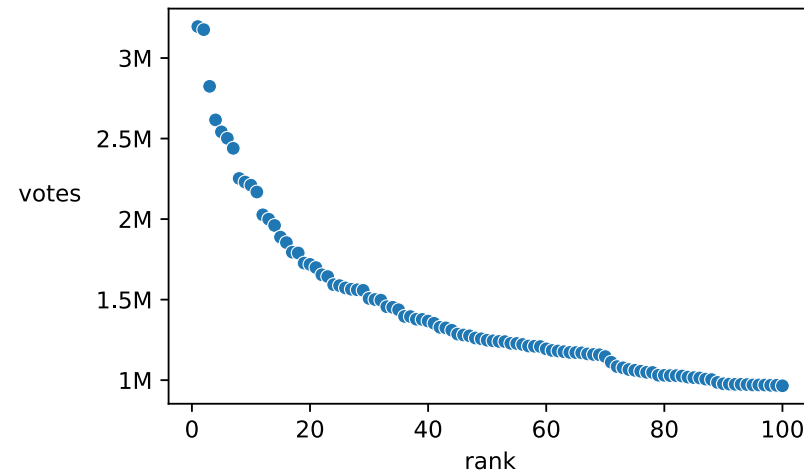
The Premise, Plotted: 100 Random Movies

```
1 imdb = pl.read_csv("../datasets/movies-from-imdb/random.csv").sample(100, seed=418)
2 imdb = imdb.sort("numVotes", descending=True).with_row_index("rank", offset=1) # rank from 1, not 0
3 ax = sns.scatterplot(imdb, x="rank", y="numVotes")
4 ax.set_ylim(bottom=0)
5 ax.set_ylabel("votes", rotation=0, labelpad=20)
6 ax.yaxis.set_major_formatter(EngFormatter(sep=""))
```



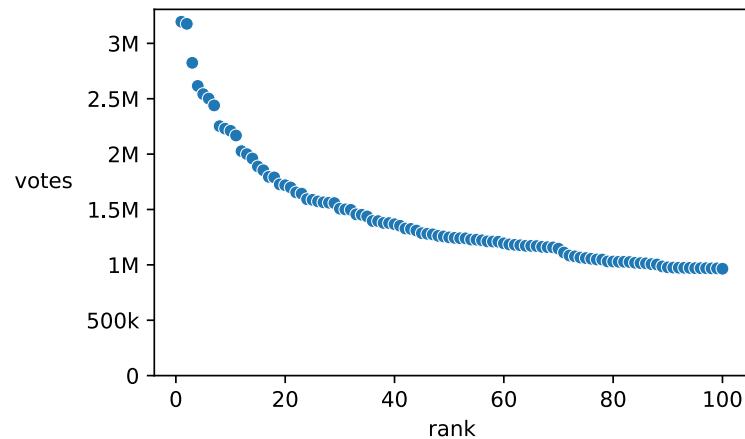
The Premise, Plotted: 100 Most Popular Movies

```
1 imdb = pl.read_csv("../datasets/movies-from-imdb/popular.csv").head(100)
2 imdb = imdb.with_row_index("rank", offset=1) # rank from 1, not 0
3 ax = sns.scatterplot(imdb, x="rank", y="numVotes")
4 ax.set_ylabel("votes", rotation=0, labelpad=20)
5 ax.yaxis.set_major_formatter(EngFormatter(sep=""))
```



The Premise, Plotted: 100 Most Popular Movies

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```



How Do Recommender Systems Work?

Basic Architectures

Example: What YouTube Knows

For every (user, video) pair, YouTube can record a **utility**:

	lofi study mix	cat fails	intro to SQL	marathon vlog	sourdough 101
Alice	95%		80%		10%
Bob		100%		60%	
Carmen	90%	15%			
Dev			85%	70%	

Consider utility as **watch percentage** (blank = never clicked/watched)

Sources

1. GitHub source: <https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week11.qmd>.
2. Recommendation systems content follows Leskovec, Rajaraman & Ullman, *Mining of Massive Datasets*, Chapter 9.
3. Linden, Smith & York, “Amazon.com recommendations: item-to-item collaborative filtering,” *IEEE Internet Computing* 7:1 (2003).
<https://doi.org/10.1109/MIC.2003.1167344>
4. Slides developed using materials from [Elena Zheleva](#) and [Gonzalo Bello Lander](#), the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
5. Slide deck built with [Quarto](#) revealjs.
6. Title font is Big Shoulders; Body font is [Libre Franklin](#).